

GALAXY CLUSTERS

Galaxies are bound by gravity to form clusters of about 100 to several thousand. Clusters vary from 3 to 35 million light-years across. Some have a concentrated central core and a well-defined spherical structure; others are irregular in shape and structure. The cluster of galaxies that contains our own galaxy is called the Local Group. The neighboring Virgo Cluster is a large, irregular

HICKSON COMPACT GROUP 87

This cluster includes a face-on spiral galaxy in the center of the image, two closer oblique spirals, and an elliptical galaxy at lower right.

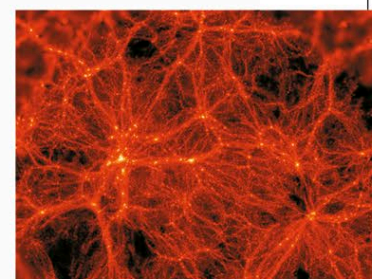
cluster of about 1,300 galaxies lying 50 million light-years away. Chains of a dozen or so galaxy clusters are linked loosely by gravity and make up superclusters, which can be 500 million light-years or more in extent. The Laniakea Supercluster, of which the Local Group is a part, is 520 million light-years long. Superclusters in turn are arranged in broad sheets and filaments, separated by voids of up to several hundred million light-years across.

RICH CLUSTER

One of the most massive galaxy clusters known, Abell 1689 is thought to contain hundreds of galaxies (colored gold here).

DARK MATTER AND DARK ENERGY

There is far more matter in the universe than is contained in stars and other visible objects. The invisible mass is called “dark matter.” Its composition is unknown. Some might take the form of MACHOs (massive compact halo objects), which include brown dwarfs and some types of black holes. However, it seems probable that the majority consists of WIMPs (weakly interacting massive particles) or some other type of as-yet undiscovered subatomic particle. Evidence that dark matter exists includes the fact that many galaxies would fly apart rather than rotate unless they contain large amounts of unseen matter. Even if all the dark matter deduced from observations is included, the density of the universe is not sufficient to satisfy theories of its evolution. To find a solution, cosmologists have proposed the existence of “dark energy,” a force that counteracts gravity and causes the universe to expand faster (see p.58). The exact nature of dark energy is still speculative.



DARK MATTER DISTRIBUTION

This image from a computer simulation shows the way in which dark matter (red clumps and filaments) must be distributed within the galaxy superclusters in our local universe.

DISTORTED GALAXY

Nicknamed “the Tadpole,” this galaxy lies 420 million light-years away. Like any galaxy, it is a vast, spinning wheel of matter bound together by gravity. In clusters, gravity can also rip galaxies apart. The streamer of stars emerging from this galaxy is thought to have been torn out by the gravity of a smaller galaxy passing close by.

EXPLORING SPACE

THE SEARCH FOR DARK MATTER

To find dark matter, scientists are investigating some of the several forms it could take. Underground detectors search for evasive particles, such as WIMPs and neutrinos. Neutrinos are so tiny, they were once thought to be massless, but they do have a minute mass. There are so many in the cosmos that their combined mass could account for about 1 percent of the universe’s dark matter. WIMPs, if detected, could account for far more.

WIMP DETECTOR

These photomultiplier tubes are part of the XENON1T detector at Gran Sasso, Italy. They function to detect light flashes as WIMPs smash into atoms in a large tank of liquid xenon.

